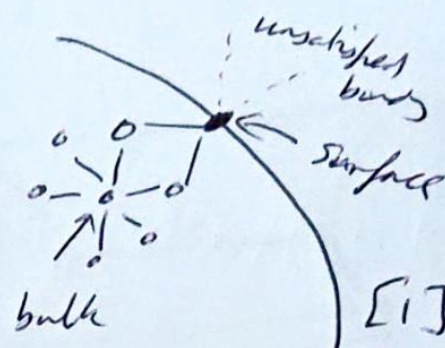


1) a) • Interactions/bonds between neighbouring molecules lower energy of system. [1]

• Each molecule forms bonds/interacts with neighbours [1]

• $N_{\text{bulk}} > N_{\text{surface}}$. [1]



• Molecules at surface have unsatisfied bonds [1]

• Energetic cost associated with molecules at surface $\rightarrow$ interfacial energy [1]

b) • Energetic cost in (a) / interfacial energy proportional to surface area [1]

② • Shape of liquid interface at equilibrium minimises exposed surface area [1]

$$c) \quad \Delta P = \gamma \left[\frac{1}{R_1} + \frac{1}{R_2} \right] \quad [1]$$

$$R_1 = -R \quad \begin{matrix} [1] \text{ for } - \\ [1/2] \text{ for } R \end{matrix}$$

$$R_2 = 0/2 \quad [1/2]$$

④

$$\Delta P \approx -\frac{\gamma}{R} \quad [1] \quad (|R| \ll 0/2)$$

1d) $P = \frac{-\gamma}{R}$ (allow carry forward for missing - in part c).

$$F = PA = \frac{P\pi D^2}{4} \quad [1]$$

$$F = -\frac{\gamma\pi D^2}{4R} \quad [1]$$

$$R^2 = \left(\frac{H}{2}\right)^2 + (R-b)^2 \quad [1]$$

$$R^2 = \frac{H^2}{4} + R^2 - 2Rb + b^2 \quad [1]$$

$$R = \frac{H^2}{8b} + \frac{b}{2} \quad [1]$$

(7) $4R = \frac{H^2}{(D-w)} + D-w$

$$F = \frac{-\gamma\pi D^2}{\frac{H^2}{D-w} + D-w} \quad \text{or}$$

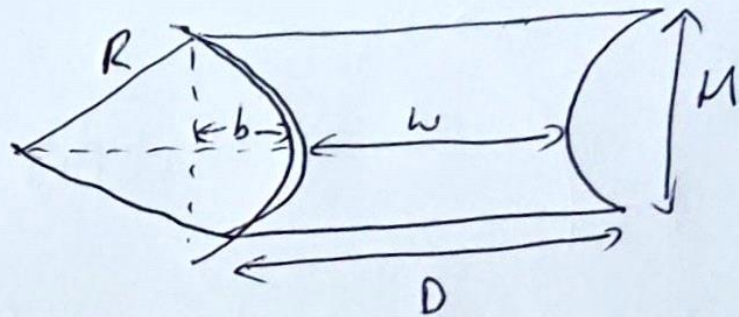
$$\frac{-\gamma\pi D^2(D-w)}{H^2 + (D-w)^2} \quad [2]$$

e) $F = -0.113 \text{ N}$ [2]

1 for number

1 for unit + sign

No carry forward for sign since should realise from diagram.



$$f) P = \gamma \left[\frac{1}{R_1} + \frac{1}{R_2} \right]$$

$$R_1 = \infty \quad R_2 = D/2 \quad [1]$$

$$P = \frac{2\gamma}{D} \quad [1]$$

$$F = \frac{2\pi\gamma D^2/4}{D} = \frac{\pi\gamma D}{2} \quad [1]$$

$$F = 0.0057 \text{ N} \quad [1]$$

2) a) i) A covalent bond arises from the sharing of a valence electron [1]

The filling of electronic shells creates a stable/energetically favourable structure [1]

ii) $\sim 100 k_B T$ [1]

b) For an atom to detach it must break free of the covalent bonds [1]

This requires a thermal fluctuation of sufficient energy [1]

The probability of a thermal fluctuation of sufficient energy is controlled by Boltzmann $P \propto e^{-U_0/k_B T}$ [1]

$P \propto e^{-100}$ is so small as to be negligible / it won't happen. [1]

c) $U(x) = -\frac{\pi n_1 \epsilon}{6} \left(\frac{1}{x^3} - \frac{1}{(x+H)^3} \right)$

↑
1 den + film

A single plane of thickness dx in cylinder probe has

$$N_2 = n_2 \pi R^2 dx \text{ atoms [1]}$$

$$U_{\text{plane-film}} = -\frac{\pi n_1 \epsilon}{6} \left(\frac{1}{x^3} - \frac{1}{(x+H)^3} \right) \times \pi R^2 dx [1]$$

2c continued.

For whole cylinder integrate

[1] limits

$$U_{\text{cylinder-film}} = -\frac{\pi^2 A_1 A_2 C}{6} R^2 \int_0^{D+L} \frac{1}{x^3} - \frac{1}{(x+H)^3} dx \quad [1]$$

$$U_{cf} = -\frac{AR^2}{6} \left[\frac{-1}{2x^2} + \frac{1}{2(x+H)^2} \right]_0^{D+L} \quad [1]$$

[1] for Hamaker

$$U_{cf} = -\frac{AR^2}{12} \left[\frac{1}{(D+H+L)^2} - \frac{1}{(D+H)^2} - \frac{1}{(D+L)^2} + \frac{1}{D^2} \right]$$

d) if $D \ll L$ some terms insignificant

$$U_{cf} \approx -\frac{AR^2}{12} \left[\frac{1}{D^2} - \frac{1}{(D+H)^2} \right] \quad [2]$$

$$e) 1) U = -\frac{5 \times 10^{-20} \times (5 \times 10^{-6})^2}{12} \left[\frac{1}{(10^{-8})^2} - \frac{1}{(10^{-8} + 0.3 \times 10^{-7})^2} \right]$$

$$= 1.04 \times 10^{-32} \left[1 \times 10^{16} - 9.43 \times 10^{15} \right]$$

$$= 5.93 \times 10^{-18} \text{ J} \quad [2]$$

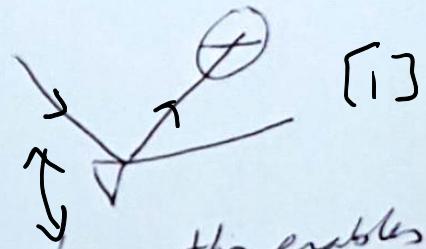
2) if $H \rightarrow \infty$ $\frac{1}{D+H} \rightarrow 0$ [1]

$$U = 1.04 \times 10^{-32} \times 10^{16} = 1.04 \times 10^{-16} \text{ J} \quad [1]$$

f). when a force interacts with the tip of an AFM this causes the cantilever to bend ~~causing the cantilever to bend~~. [1]

• The deflection is measured using a laser which reflects off back of cantilever onto split photodiode [1]

• Diagram showing deflection of cantilever, laser & PD. [1]



• If the spring constant of cantilever is known this enables one to measure the force. [1] $F = kZ$

• The spring constant can be measured from displacements caused by thermal noise/fluctuations [1]

• $K = \frac{k_B T}{\langle Z^2 \rangle}$ [1]

3a) 1) - A colloid in a liquid will undergo Brownian motion [1]
- This results in a random walk [1]

$$\left. \begin{aligned} 2) \quad \langle R^2 \rangle &\sim 6Dt \\ D &= \frac{kT}{6\pi\eta a} \end{aligned} \right\} [1]$$

$$\langle R^2 \rangle \sim \frac{kTt}{\pi\eta a} [1]$$

$$t = \frac{4\pi\eta a}{kT d^2} [1]$$

3) - The colloid would visit all points in the cavity [1]
- Each point would be visited with equal probability [1]

b) - A microstate is a particular configuration of the particles within the cavity (positions/velocities). [1]

- The entropy is directly related to the number of accessible microstates [1]

$$- S = k_B \ln W. [1]$$

c) - When the large colloid comes close to wall the small spheres are excluded from the gap. [1]

- However, The total free volume available to microspheres increases. [1]

- Number of microstates increases [1]

- Entropy increases [1]

- $F = +T \frac{dS}{dD} \rightarrow$ Force attracting big sphere to wall preventing it diffusing out of cavity [1]

$$3d) \quad k = \frac{k_B T}{\langle x^2 \rangle} \quad [1]$$

$$k = \frac{1.38 \times 10^{-23} \times 290}{5 \times 10^{-18}}$$

$$k = 8 \times 10^{-4} \text{ Nm}^{-1} \quad [1]$$

3e)

$$U = -\frac{n\pi kT}{3} [a+2r-D]^2 [2a+r+D]$$

$$F = -\frac{dU}{dD} \quad [1]$$

$$F = \frac{n\pi kT}{3} \frac{d}{dD} \underbrace{(a+2r-D)^2}_{x} \underbrace{(2a+r+D)}_y$$

By product rule $\frac{dU}{dD} = x \frac{dy}{dD} + y \frac{dx}{dD}$

$$\frac{dx}{dD} = -2(a+2r-D) \quad [1]$$

$$\frac{dy}{dD} = 1$$

$$F = \frac{n\pi kT}{3} \left((a+2r-D)^2 + (2a+r+D) \times -2(a+2r-D) \right)$$

$$F = \frac{n\pi kT}{3} \left((a+2r-D) \left((a+2r-D) - (4a+2r-2D) \right) \right) \quad [1]$$

$$F = -\frac{n\pi kT}{3} (a+2r-D)(a+D) \quad [1]$$

valid for $a+2r > D \quad [1]$

3f)

$$F_{\text{trap}} = F_{\text{depletion}} \quad [1]$$

$$\Delta r = \frac{F_{\text{depletion}}}{k}$$

$$F_{\text{depletion}} = -10^{20} \times \pi \times 1.38 \times 10^{-23} \times 290 \times (5 \times 10^{-6} + 2 \times 50 \times 10^{-9} - 5.05 \times 10^{-6}) \\ \times (5 \times 10^{-6} + 5.05 \times 10^{-6})$$

$$F_{\text{depletion}} = 0.63 \text{ pN} \quad [1]$$

$$\Delta r = \frac{6.3 \times 10^{-13}}{8 \times 10^{-4}} \quad \leftarrow \text{from part d}$$

$$\underline{\underline{\Delta r = 0.79 \text{ nm}}} \quad [1]$$